

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java
Practical - VI

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch: 2
Date of Assignment: 22-8-26	Date/Time of Submission: 22-8-26

5. Implement the following Intermediate JSP Applications.

a. Develop a JSP application to demonstrate the use of JSP Directives and Implicit Objects.

CODE :-

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.Date" %>
<html>
<head>
<title>JSP Directives and Implicit Objects</title>
</head>
<body>
<h2>JSP Directives and Implicit Objects</h2>
<h3>JSP Directives</h3>
<p>Language: Java</p>
<p>Content Type: <%= response.getContentType() %></p>
<p>Current Date: <%= new Date() %></p>
<h3>Implicit Objects</h3>
<p>1. Request Method: <%= request.getMethod() %></p>
<p>2. Response Content Type: <%= response.getContentType() %></p>
<p>3. Session ID: <%= session.getId() %></p>
<p>4. Application Server: <%= application.getServerInfo() %></p>
<p>5. Page Name: <%= page.getClass().getName() %></p>
<p>6. PageContext: <%= pageContext.getClass().getName() %></p>
<p>7. Request Parameter: <%= request.getParameter("name") %></p>
<p>8. Config Servlet Name: <%= config.getServletName() %></p>
<p>9. Exception Object: Exception object is available only in error pages.</p>
</body>
</html>
```

OUTPUT: -

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %> <%@ page import="java.util.Date" %>
```

JSP Directives and Implicit Objects

JSP Directives

Language: Java

Content Type: <%= response.getContentType() %>

Current Date: <%= new Date() %>

Implicit Objects

1. Request Method: <%= request.getMethod() %>

2. Response Content Type: <%= response.getContentType() %>

3. Session ID: <%= session.getId() %>

4. Application Server: <%= application.getServerInfo() %>

5. Page Name: <%= page.getClass().getName() %>

6. PageContext: <%= pageContext.getClass().getName() %>

7. Request Parameter: <%= request.getParameter("name") %>

8. Config Servlet Name: <%= config.getServletName() %>

9. Exception Object: Exception object is available only in error pages.

b. Develop a JSP application to demonstrate Form Processing and Input Validation.**CODE:-****admission.jsp**

```

<html>
<head>
<title>Student Admission Form</title>
</head>
<body>
<h2>Student Admission Form</h2>
<form action="process.jsp" method="post">
Name:
<input type="text" name="name">
<br><br>

Age:
<input type="text" name="age">
<br><br>

Email:
<input type="text" name="email">
<br><br>

Mobile:
<input type="text" name="mobile">
<br><br>

Gender:
<input type="radio" name="gender" value="Male"> Male
<input type="radio" name="gender" value="Female"> Female
<br><br>

Course:
<select name="course">
<option value="">Select Course</option>
<option value="BSc IT">BSc IT</option>
<option value="BCA">BCA</option>
<option value="MSc IT">MSc IT</option>
<option value="MCA">MCA</option>
</select>
<br><br>

Hobbies:
<input type="checkbox" name="hobby" value="Reading"> Reading
<input type="checkbox" name="hobby" value="Sports"> Sports
<input type="checkbox" name="hobby" value="Music"> Music
<br><br>

Address:
<textarea name="address"></textarea>
<br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>

```

process.jsp

```

<html>
<head>
<title>Admission Details</title>
</head>

```

```

<body>
<h2>Admission Details</h2>
<%
String name = request.getParameter("name");
String age = request.getParameter("age");
String email = request.getParameter("email");
String mobile = request.getParameter("mobile");
String gender = request.getParameter("gender");
String course = request.getParameter("course");
String address = request.getParameter("address");
String[] hobbies = request.getParameterValues("hobby");
boolean valid = true;
if(name == null || name.trim().equals(""))
{
out.println("Name is required.<br>");
valid = false;
}
if(age == null || age.trim().equals(""))
{
out.println("Age is required.<br>");
valid = false;
}
if(email == null || email.trim().equals(""))
{
out.println("Email is required.<br>");
valid = false;
}
if(mobile == null || mobile.trim().equals(""))
{
out.println("Mobile number is required.<br>");
valid = false;
}

if(gender == null)
{
out.println("Please select gender.<br>");
valid = false;
}
if(course == null || course.equals(""))
{
out.println("Please select a course.<br>");
valid = false;
}
if(hobbies == null)
{
out.println("Please select at least one hobby.<br>");
valid = false;
}
if(address == null || address.trim().equals(""))
{
out.println("Address is required.<br>");
valid = false;
}
if(valid)

```

```

{
%>
<h3>Student Admission Successful</h3>
Name: <%= name %><br><br>
Age: <%= age %><br><br>
Email: <%= email %><br><br>
Mobile: <%= mobile %><br><br>
Gender: <%= gender %><br><br>
Course: <%= course %><br><br>
Hobbies:
<%
for(String hobby : hobbies)
{
out.println(hobby + " ");

}
%>
<br><br>
Address: <%= address %><br><br>
<%
}
%>
</body>
</html>

```

OUTPUT:-

localhost:8080/Practical6/admission.jsp

Run it in the local host adding admission.jsp

Student Admission Form

Name:

Age:

Email:

Mobile:

Gender: ☐ Male ☒ Female

Course:

Hobbies: ☐ Reading ☐ Sports ☒ Music

Address:

Admission Details

Student Admission Successful

Name: Khushi Patel

Age: 19

Email: khushipatelmvl@gmail.com

Mobile: 7045564157

Gender: Female

Course: BSc IT

Hobbies: Music

Address: Siddharth Nagar, Santacruz E

c. Develop a JSP application to display data from a MySQL Database using JDBC.**CODE:-****Sql code :**

```

CREATE DATABASE ecommerce;
USE ecommerce;
Database changed
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    brand VARCHAR(50),
    price DECIMAL(10,2),
    quantity INT,
    rating DECIMAL(2,1),
    discount DECIMAL(5,2),
    description VARCHAR(255)
);

```

Index.jsp :

```

<%@ page language="java"
    contentType="text/html; charset=UTF-8"
    import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>E-Commerce Products</title>
<style>
    body {
        font-family: Arial, sans-serif;
        background-color: #f2f2f2;
        margin: 30px;
    }
    h2 {
        text-align: center;
        color: #333;
    }
    table {
        width: 100%;
        border-collapse: collapse;
        background-color: white;
    }
    th {
        background-color: #2874A6;
        color: white;
        padding: 12px;
    }
    td {
        padding: 10px;
        border: 1px solid #ddd;
        text-align: center;
    }
    tr:nth-child(even) {
        background-color: #f2f2f2;
    }
    tr:hover {
        background-color: #D6EAF8;
    }

```

```

</style>
</head>
<body>
<h2>E-Commerce Product List</h2>
<table>
  <tr>
    <th>Product ID</th>
    <th>Product Name</th>
    <th>Category</th>
    <th>Brand</th>
    <th>Price</th>
    <th>Quantity</th>
    <th>Rating</th>
    <th>Discount</th>
    <th>Description</th>
  </tr>
  <%
    Connection con = null;
    Statement stmt = null;
    ResultSet rs = null;
    try {
      Class.forName("com.mysql.jdbc.Driver");
      con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/ecommerce",
        "root",
        "root"
      );
      stmt = con.createStatement();
      rs = stmt.executeQuery("SELECT * FROM product");
      while (rs.next()) {
  %>
  <tr>
    <td><%= rs.getInt("product_id")%></td>
    <td><%= rs.getString("product_name")%></td>
    <td><%= rs.getString("category")%></td>
    <td><%= rs.getString("brand")%></td>
    <td>₹<%= rs.getDouble("price")%></td>
    <td><%= rs.getInt("quantity")%></td>
    <td><%= rs.getDouble("rating")%>  </td>
    <td><%= rs.getDouble("discount")%>%</td>
    <td><%= rs.getString("description")%></td>
  </tr>
  <%
    }

    } catch (Exception e) {
      out.println("<p>Error: " + e.getMessage() + "</p>");
    } finally {

      try {
        if (rs != null) {
          rs.close();
        }
        if (stmt != null) {
          stmt.close();
        }
      }
    }
  %>

```

```

        if (con != null) {
            con.close();
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
}
%>
</table>
</body>
</html>

```

OUTPUT:-

```

mysql> SELECT * FROM product;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| product_id | product_name | category | brand | price | quantity | rating | discount | description |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | iPhone 15 | Mobile | Apple | 69999.00 | 10 | 4.8 | 10.00 | Apple iPhone with advanced camera and performance |
| 2 | Galaxy S24 | Mobile | Samsung | 64999.00 | 15 | 4.7 | 12.00 | Samsung smartphone with AMOLED display |
| 3 | MacBook Air M2 | Laptop | Apple | 89999.00 | 8 | 4.9 | 8.00 | Lightweight laptop powered by Apple M2 chip |
| 4 | WH-1000XM5 | Headphones | Sony | 29999.00 | 20 | 4.6 | 15.00 | Wireless noise cancelling headphones |
| 5 | Inspiron 15 | Laptop | Dell | 55999.00 | 12 | 4.5 | 10.00 | Dell laptop suitable for everyday computing |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

E-Commerce Product List

Product ID	Product Name	Category	Brand	Price	Quantity	Rating	Discount	Description
1	iPhone 15	Mobile	Apple	₹69999.0	10	4.8 ★	10.0%	Apple iPhone with advanced camera and performance
2	Galaxy S24	Mobile	Samsung	₹64999.0	15	4.7 ★	12.0%	Samsung smartphone with AMOLED display
3	MacBook Air M2	Laptop	Apple	₹89999.0	8	4.9 ★	8.0%	Lightweight laptop powered by Apple M2 chip
4	WH-1000XM5	Headphones	Sony	₹29999.0	20	4.6 ★	15.0%	Wireless noise cancelling headphones
5	Inspiron 15	Laptop	Dell	₹55999.0	12	4.5 ★	10.0%	Dell laptop suitable for everyday computing